

THE SARMANOV FAMILY AND ITS GENERALIZATION

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Summary: A general class of bivariate distributions is introduced. This class includes the so-called Sarmanov-Lee class (and consequently the Farlie-Gumbel-Morgenstern class). It is shown that using procedures described in the paper it is possible to construct distributions of the FGM form for which the correlation coefficient between the marginals can achieve values close to ± 0.6 .

1. Introduction

The assumption of independent random variables is not always valid and viable in statistical applications. System reliability, representation of systems, dependence bound stochastic processes such as spatial processes and risk analysis are some application areas, among many others, that can be mentioned in this respect. There are many concepts and modelling approaches about the modelling of dependence. Multivariate distribution theory sets the fundamental ground regarding dependence modelling. In this sense the

classical bivariate Farlie-Gumbel-Morgenstern (FGM) system of distributions constitutes a class of multivariate distributions with given marginals. The FGM class of distributions, originally introduced by Morgenstern (1956), investigated by Gumbel (1960), for exponential marginals and further generalized by Farlie (1960) is important and efficient in applications. The FGM system has the following joint cumulative distribution function form of the underlying variables (X, Y)

$$F(x, y) = F_X(x)G_Y(y) \{1 + \alpha \bar{F}_X(x)\bar{G}_Y(y)\}, \quad (1.1)$$

where F_X and G_Y are the marginal cumulative distributions, $\bar{F}_X = 1 - F_X$, $\bar{G}_Y = 1 - G_Y$ and α is a real number restricted between 1 and -1 so that $F(x, y)$ is a distribution function. The probability density function (p.d.f.) of the FGM system is

$$f(x, y) = f(x)g(y) \{1 + \alpha(1 - 2F_X(x)(1 - 2G_Y(y))\}, \quad (1.2)$$

where f and g are corresponding p.d.f.'s. A general FGM system construction has the copula

$$C_\alpha(xy) = xy(1 + \alpha A(x)B(y)),$$

where the pair of kernels $A(x)$ and $B(y)$ satisfy certain regularity conditions ensuring that $C_\alpha(x, y)$ is a copula. Note that a two-dimensional copula is a function $C(x, y)$ from $[0, 1] \times [0, 1]$ to $[0, 1]$ with the properties:

1. $C(x, 0) = 0 = C(0, y)$, $C(x, 1) = x$ and $C(1, y) = y$;
2. For every x_1, x_2, y_1, y_2 such that $0 \leq x_1 < x_2 \leq 1$ and $0 \leq y_1 < y_2 \leq 1$

$$C(x_2, y_2) - C(x_2, y_1) - C(x_1, y_2) + C(x_1, y_1) \geq 0.$$

According to Sklar's Theorem if $F(x, y)$ is a joint distribution function with continuous margins $F_X(x)$ and $F_Y(y)$, then there exists a unique copula C such that $F(x, y) = C(F_X(x), F_Y(y))$. Theory and applications of copulas are well documented in Nelsen (1998).

It is well known that the correlation coefficient of FGM distributions lies between $-1/3$ and $1/3$ (see e.g. Johnson and Kotz, 1975). Huang and Kotz (1984) determine the natural parameter space of an iterated FGM distribution proposed by Johnson and Kotz (1977), and show that the range of correlation can be widened. They also show that there exist no marginals for which a single iteration will bring about higher negative correlation. Huang and Kotz (1999) extended the bivariate FGM distribution in their attempts to increase the dependence between the underlying variables by introducing an additional parameter. Bairamov, Kotz and Bekçi (2001) provide modification of the FGM distribution which allows us to achieve correlation between the components to be greater than $1/2$, and study distributional properties of concomitants for this generalized FGM distribution.

The class of bivariate distributions introduced by Sarmanov (1966) encompasses the FGM family. Specifically, let f and g be univariate p.d.f.'s. Let $\psi_i(t)$, $i = 1, 2$, be bounded nonconstant functions such that

$$\int_{-\infty}^{\infty} f(t)\psi_1(t)dt = 0 \quad \text{and} \quad \int_{-\infty}^{\infty} g(t)\psi_2(t)dt = 0.$$

Define a bivariate density function

$$h_{\alpha}(x, y) = f(x)g(y) \{1 + \alpha\psi_1(x)\psi_2(y)\} \quad (1.3)$$

with specified marginals $f(x)$ and $g(x)$, where α is a real number satisfying the condition that $1 + \alpha\psi_1(x)\psi_2(y) \geq 0$ for all x and y . The family with joint p.d.f. (1.3) is called the Sarmanov family of bivariate distributions. Lee

(1996) considers “kernels” of type $\psi_1(x) = x - \mu_1$, $\psi_2(y) = y - \mu_2$, where $\mu_1 = E(X)$, $\mu_2 = E(Y)$, and shows that the range of correlation coefficients for this family of distributions is determined by both the marginal distributions and their mixing functions ψ_i ($i = 1, 2$) and may therefore be wider than the one for the FGM distributions. The range of α in this case is:

$$\max\left(\frac{-1}{\mu_1\mu_2}, \frac{-1}{(1-\mu_1)(1-\mu_2)}\right) \leq \alpha \leq \min\left(\frac{1}{\mu_1(1-\mu_2)}, \frac{1}{\mu_2(1-\mu_1)}\right) \quad (1.4)$$

and the correlation coefficient is given by

$$\rho = \frac{\alpha\nu_1\nu_2}{\sigma_X\sigma_Y}, \quad (1.5)$$

where $\nu_1 = \int x\psi_1(x)f(x)dx$, $\nu_2 = \int y\psi_2(y)g(y)dy$ and $\sigma_X = \sqrt{\text{Var}(X)}$, $\sigma_Y = \sqrt{\text{Var}(Y)}$. However, for the *uniform* $[0, 1]$ marginals it follows from (1.4) and (1.5) that

$$-4 \leq \alpha \leq 4$$

and

$$-\frac{1}{3} \leq \rho \leq \frac{1}{3},$$

which is the same range as for classical FGM.

The uniform representation of the FGM distribution leads to families of distributions with fixed marginals. Lai and Xie (2000) present a generalization of the FGM family with uniform marginals that possesses positive quadrant dependence. In addition to such distributional characterizations, measurement of dependence by correlation coefficient, as a very important application tool, is yet to be explored for further improvements.

Motivated by Lee's (1996) paper we present herein some constructions which generalize the FGM and Sarmanov-Lee models. Using these constructions leads us to the result that even in the case of uniform marginals the correlation coefficient can achieve large values. Thus the popular opinion

that bivariate distributions of the FGM form yield low values of the correlation coefficient between the marginals has been refuted:

2. A generalization of the FGM and Sarmanov class of distributions

Consider the following class of bivariate distributions with absolutely continuous marginals F, G given by the joint density $f(x, y)$:

$$f(x, y) = f(x)g(y) \{1 + \alpha H(F(x), G(y))\}, \quad (2.1)$$

where $f(x), g(y)$ are specified marginal densities and $H(x, y)$ is an integrable function defined on $[0, 1]$ satisfying

$$\int_0^1 H(x, y) dx = 0, \quad \int_0^1 H(x, y) dy = 0, \quad (2.2)$$

and α is a real number satisfying the condition that $1 + \alpha H(x, y) \geq 0$ for all $0 \leq x, y \leq 1$,

Denote by $D^+ = \{(x, y) : 0 \leq x, y \leq 1, H(x, y) > 0\}$, and by $D^- = \{(x, y) : 0 \leq x, y \leq 1, H(x, y) < 0\}$. The admissible range for α ensuring that (2.1) is a joint density can be expressed in terms of $\max_{(x, y) \in D^+} (H(x, y))$ and $\max_{(x, y) \in D^-} (-H(x, y))$ as follows:

$$-\frac{1}{\max_{(x, y) \in D^+} H(x, y)} \leq \alpha \leq \frac{1}{\max_{(x, y) \in D^-} (-H(x, y))}.$$

Observe that $H(x, y) = (1 - 2x)(1 - 2y)$ corresponds to the one parameter FGM family, if $H(x, y) = A(x)B(y)$ with A, B satisfying certain conditions one has the extended FGM family. Specifically if $H(x, y) = (1 - (1 + p)x^p)(1 - (1 + p)y^p)$, $p > 0$ or $H(x, y) = (1 - x)^{p-1}(1 - y)^{p-1}(1 - (1 + p)x)(1 - (1 + p)y)$, $p > 1$ one has

respectively the type 1 and type 2 Huang-Kotz FGM families of joint distributions (see Huang and Kotz, 1999). Furthermore, if $H(x, y) = \varphi_1(F^{-1}(x))\varphi_2(G^{-1}(y))$, where $\varphi_1(x)$ and $\varphi_2(x)$ are bounded, nonconstant functions such that

$$\int_{-\infty}^{\infty} \varphi_1(x)f(x)dx = 0 \text{ and } \int_{-\infty}^{\infty} \varphi_2(x)g(x)dx = 0,$$

one has the Sarmanov family of bivariate densities.

Intuitively the most reasonable form for selecting H (in the spirit of the original FGM distribution) is

$$H(x, y) = \varphi_1(x)\varphi_2(y), \quad (2.3)$$

where $\varphi_1(x)$ and $\varphi_2(y)$ satisfy

$$\int_0^1 \varphi_1(x)dx = 0 \text{ and } \int_0^1 \varphi_2(y)dy = 0.$$

This case is equivalent to the Sarmanov construction. Additional choices for $H(x, y)$ may be, for example:

1. $H(x, y) = \left[x^p \ell n \frac{1}{x} - \frac{1}{(p+1)^x} \right] \left[y^p \ell n \frac{1}{y} - \frac{1}{(p+1)^y} \right]$, where $p > 0$.
2. $H(x, y) = \left[(1 - x^n)^{p-1} (1 - (1 + np) x^n) \right] \cdot \left[(1 - y^n)^{p-1} (1 - (1 + np) y^n) \right]$, $p > 0$.
3. $H(x, y) = \left[\left(\ell n \frac{1}{x} \right)^q - \Gamma(q+1) \right] \left[\left(\ell n \frac{1}{y} \right)^q - \Gamma(q+1) \right]$, $q > 0$ etc., where $\Gamma(q)$ is the gamma function.

Now we provide an example of $H(x, y)$ which satisfies (2.2) but does not fit into the Sarmanov-Lee construction.

Example 2.1. Let $\varphi(x)$ be continuous on the $[0, 1]$ function. Define

$$H_{\varphi}(x, y) = \begin{cases} \{\varphi(x) - \varphi(1/2 - x)\} \{\varphi(y) - \varphi(1/2 - y)\} & \text{if } 0 < x, y < 1/2 \\ \{\varphi(x) - \varphi(3/2 - x)\} \{\varphi(y) - \varphi(3/2 - y)\} & \text{if } 1/2 < x, y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Since

$$\int_0^1 \varphi(x) dx = \int_0^1 \varphi(1-x) dx. \quad (2.4)$$

Here H_{φ} satisfies the conditions (2.2), but evidently not (2.3). For the particular choice $\varphi(x) = x - \frac{1}{2}$ the range of α is

$$-4 \leq \alpha \leq 4.$$

We now provide some properties of the family given by (2.1).

1. $E(XY) = \mu_X \mu_Y + \alpha \int_0^1 \int_0^1 F^{-1}(x) G^{-1}(y) H(x, y) dx dy$, where $F^{-1}(x) = \inf \{y : F(y) \geq x\}$ is the inverse function of F and $\mu_X = E(X)$, $\mu_Y = E(Y)$.
2. $cov(X, Y) = \alpha \int_0^1 \int_0^1 F^{-1}(x) G^{-1}(y) H(x, y) dx dy$.
3. $P\{X \leq x | Y = y\} = F(x) + \alpha \int_{-\infty}^x H(F(u), G(y)) dF(u)$.
4. $E(X | Y = y) = \mu_X + \alpha \int_0^1 F^{-1}(x) H(x, G(y)) dx$, namely if $H(x, y)$ is monotone with respect to y then X and Y are regression dependent.
5. The correlation coefficient of X and Y , if it exists, is given by

$$\rho = \frac{\alpha}{\sigma_X \sigma_Y} \int_0^1 \int_0^1 F^{-1}(x) G^{-1}(y) H(x, y) dx dy.$$

Therefore in family (2.1), like the "normal" case, the correlation coefficient ρ totally governs the dependence. Using Schwarz inequality, we have

$$|\rho| \leq |\alpha| \left(1 + \frac{\mu_X^2}{\sigma_X^2}\right)^{\frac{1}{2}} \left(1 + \frac{\mu_Y^2}{\sigma_Y^2}\right)^{\frac{1}{2}} \left(\int_0^1 \int_0^1 H^2(x, y) dx dy\right)^{\frac{1}{2}}.$$

3. A subclass of Sarmanov family

Let $\varphi(x)$, $0 \leq x \leq 1$ be a continuous function and denote $A(x) = \varphi(x) - \varphi(1-x)$. Consider a class of bivariate distributions with p.d.f. of the form

$$f_\alpha(x, y) = 1 + \alpha A(x)A(y), \quad 0 \leq x, y \leq 1. \quad (3.1)$$

In view of (2.5) $f_\alpha(x, y)$ is a joint p.d.f. with uniform marginals. Note that (3.1) is a subclass of the Sarmanov family. The admissible range for α for the class (3.1) can be expressed in terms of $\max_{D^+} A(x)A(y)$ and $\max_{D^-} (-A(x)A(y))$ as follows:

$$-\frac{1}{\max_{D^+} A(x)A(y)} \leq \alpha \leq \frac{1}{\max_{D^-} (-A(x)A(y))}, \quad (3.2)$$

where $D^+ = \{(x, y) : A(x)A(y) \geq 0\}$, $D^- = \{(x, y) : A(x)A(y) \leq 0\}$. We shall call $A(x) = \varphi(x) - \varphi(1-x)$ the kernel and the function $\varphi(x)$ will be referred to as a basis function. Graphs for kernels with power basis functions $\varphi(x) = x^{1.4}$, x^4 , x^{10} are given below:

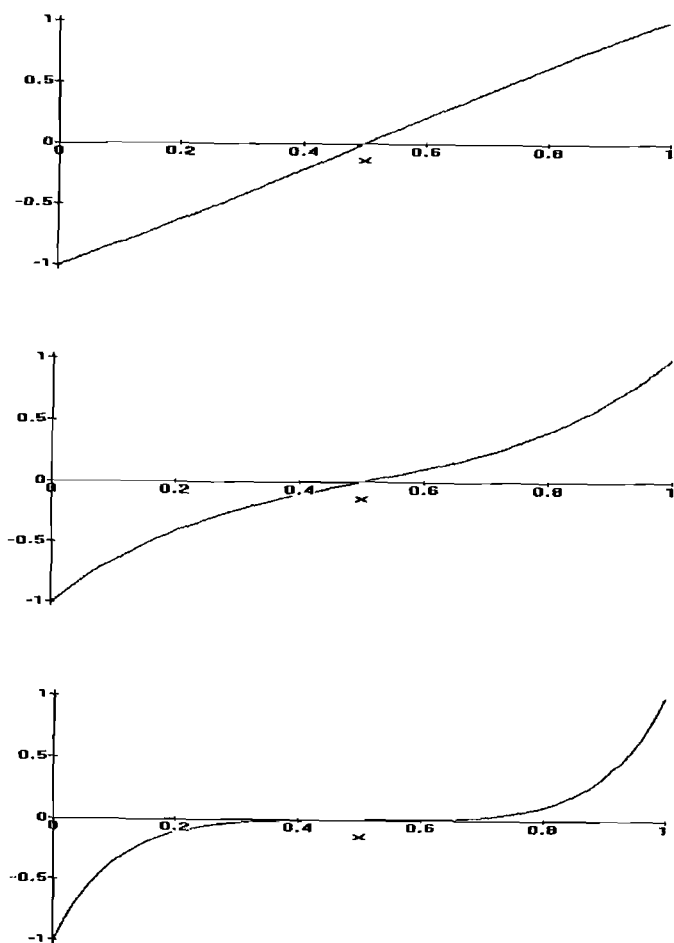


Figure 1. Graphs of kernels $A(x) = x^n - (1-x)^n$ for $n = 1.4, 4, 10$

Evidently the admissible range for α is $-1 \leq \alpha \leq 1$, this can be observed by referring to Figure 1. Using symmetry, (3.2) can also be written as follows:

$$-\frac{1}{\left\{\max_{0 \leq x \leq 1} A(x)\right\}^2} \leq \alpha \leq \frac{1}{\left\{\max_{0 \leq x \leq 1} A(x)\right\}^2}. \quad (3.3)$$

Inequalities (3.3) are convenient for calculations of the admissible range of α . The correlation coefficient ρ between the marginal variables of the joint density given by (3.1) is

$$\rho = 12\alpha \left[\int_0^1 A(x)x dx \right]^2. \quad (3.4)$$

For $\varphi(x) = x^n$, $n \geq 1$ one has

$$\int_0^1 x A(x) dx = \int_0^1 (x^n - (1-x)^n) x dx,$$

hence

$$\rho = 12\alpha \left\{ \frac{1}{n+2} - \text{Beta}(2, n+1) \right\}^2 = 12\alpha \left\{ \frac{n}{(n+1)(n+2)} \right\}^2, \quad (1)$$

where $\text{Beta}(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt$. Specifically, $\rho = \alpha/3$ for $n = 1, 2$; $\rho = 0.27\alpha$ for $n = 3$, etc. Consider (3.5) for $\alpha = 1$ and let us investigate the extreme of the function $\rho(n) = 12 \left\{ \frac{n}{(n+1)(n+2)} \right\}^2$. It is easy to verify that the solution of the equation $\rho'(n) = 0$, i.e. the extreme point of $\rho(n)$ is $n^* = \sqrt{2}$ and at this point $\rho(n)$ has a maximum. Thus for $n = \sqrt{2}$ one has $\rho_{\max} = \frac{12 \times 2}{(4 \times 3 \sqrt{2})^2} \cong 0.353249 > 1/3$ (and

$\rho_{\min} \cong -0.353249$). Below, we provide the graphs of the maximum and minimum correlation coefficient for $1 \leq n \leq 10$.

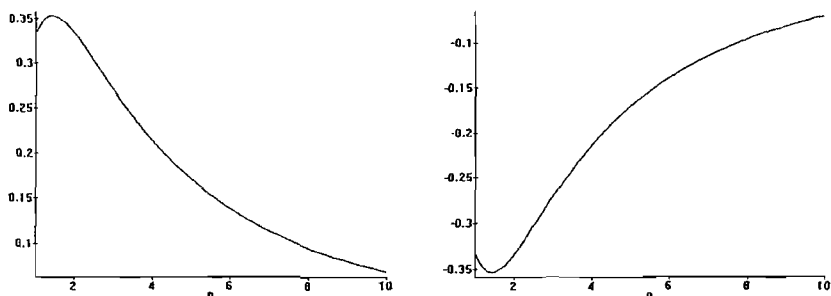


Figure 2. Graph of $\rho_{\max}$ and $\rho_{\min}$ for $1 \leq n \leq 10$

Analyzing formula (3.4) one observes that $\int_0^1 xA(x)dx$ plays an important role as far as values of the correlation coefficient ρ are concerned for the mode under consideration. To achieve large values of the correlation coefficient among the kernels having the same admissible range for α , one must select those kernels which have maximal value for $\int_0^1 xA(x)dx$.

3.1 Kernel with basis function of the type: $\varphi(x) = e^{-nx}x^p$, $p > 1$

Consider now the basis function $\varphi(x) = e^{-nx}x^p$, where n and p are positive real numbers. Then the kernel $A(x) = \varphi(x) - \varphi(1-x) = e^{-nx}x^p - e^{-n(1-x)}(1-x)^p$ and

$$A'(x) = e^{-nx}x^{p-1}(p - nx) + e^{-n(1-x)}(1-x)^{p-1}(p - n(1-x)).$$

Since $A'(x) > 0$ for $p \geq n$ and $0 \leq x \leq 1$ the function $A(x)$ increases in $[0, 1]$. The maximum value $\max A(x) = e^{-n}$ is attained at the point $x = 1$, the minimum value $\min A(x) = -e^{-n}$ is attained at the point $x = 0$.

Let $p \geq n$. Then using (3.3) one obtains the admissible range of α given by:

$$-e^{2n} \leq \alpha \leq e^{2n}.$$

Using (3.4) we have the following expression for the correlation coefficient:

$$\rho = 12\alpha \left\{ \int_0^1 x \left[e^{-nx} x^p - e^{-n(1-x)} (1-x)^p \right] dx \right\}^2.$$

Denote by $a_* = \max_{0 \leq x \leq 1} A(x)$. Then $\alpha_{\max} = a_*^{-2}$ and $\alpha_{\min} = -a_*^{-2}$.

Some selected values for

$$\rho_{\max} = 12a_*^{-2} \left\{ \int_0^1 x (e^{-nx} x^p - e^{-n(1-x)} (1-x)^p) dx \right\}^2$$

are given below:

(For $p \geq n$: $a_* = e^{-n}$. For $n = 3, p = 2$: $a_* = 0.05$; for $n = 4, p = 3$: $a_* = 0.02$; for $n = 5, p = 3$: $a_* = 0.008$; for $n = 6, p = 3$: $a_* = 0.003$.)

n	1	1.5	2	2	3	3	4	5	6	4	5	6
p	1.2	2	1.8	2	2	3	3	3	3	5	5	6
$\rho_{\max}$	0.41	0.43	0.43	0.45	0.44	0.52	0.51	0.58	0.31	0.42	0.54	0.54

The pair $(n, p) = (5, 3)$ provides a maximal value for ρ exceeding those obtained by the Huang-Kotz classes (see Huang-Kotz, 1999).

3.2 Kernel with basis function of the type: $\varphi(x) = x^p e^{-nx^q}$

Consider now the kernel with basis function $\varphi(x) = x^p e^{-nx^q}$, where n, p and q are positive real numbers. Then $A(x) = x^p e^{-nx^q} - (1-x)^p e^{-n(1-x)^q}$.

Denote $a_* = \max_{0 \leq x \leq 1} A(x)$. Then $\alpha_{\max} = a_*^{-2}$ and $\alpha_{\min} = -a_*^{-2}$.

Below are some selected values for

$$\rho_{\max} = 12a_*^{-2} \left\{ \int_0^1 x (x^p e^{-nx^q} - (1-x)^p e^{-n(1-x)^q}) dx \right\}^2 :$$

n	1	1	1	1	1.5	2	2	3	2	2	2
p	1	1	1	1	1.5	2	3	3	4	100	10
q	2	4	10	100	10	2	2	3	2	100	2
a_*	0.38	0.37	0.55	0.90	0.53	0.14	0.14	0.1	0.14	0.82	0.06
$\rho_{\max}$	0.35	0.56	0.51	0.36	0.45	0.57	0.59	0.5	0.46	0.26	0.43

3.3 Some dependence properties

We have thus constructed bivariate distributions of an extended FGM form which yields the value of the correlation near 0.6, a long way from the original value $\rho = \frac{1}{3}$. The problem of choosing an optimum kernel seems to be unsolved as yet.

The main justification for the interest in this family is the constructive approach involved in its definition and the transparent manner in which the dependence is created. Although the dependence is expressed here solely in terms of the correlation coefficient it can be shown that the family (for $\rho > 0$) possesses the TP_2 property which is the strongest among the dependence properties.

The pair of variables (X, Y) is said to be totally positively dependent (TP_2) if their joint density $f(x, y)$ satisfies the following condition: for all $x_1 < x_2$ and $y_1 < y_2$

$$f(x_1, y_1)f(x_2, y_2) - f(x_2, y_1)f(x_1, y_2) \geq 0.$$

It is well known that (for details on dependence properties (see e.g., Barlow and Proschan, 1981, Joe, 1990) the TP_2 property implies the following important dependence properties:

1. Stochastically increasing positive dependence- $SI(X|Y)$ (X is stochastically increasing in Y if $P\{X > x | Y = y\}$ is a nondecreasing function of y for all x);
2. Right-tail increasing - $RTI(X|Y)$ (X is right-tail increasing in Y if $P\{X > x | Y > y\}$ increases in y for all x);
3. $A(X, Y)$ -association (the random vector (X, Y) is associated if the inequality $E[g_1(X, Y)g_2(X, Y)] \geq E[g_1(X, Y)]E[g_2(X, Y)]$ holds for all real-valued functions g_1, g_2 which are increasing in each component and such that the expectations exist);
4. Positive quadrant dependence- PQD (the random vector (X, Y) is positive quadrant dependent if $P\{X \leq x, Y \leq y\} \geq P\{X \leq x\}P\{Y \leq y\}$).

Consider now a class of bivariate distributions with p.d.f. of the form $f(x, y) = 1 + \alpha A(x)A(y)$, $0 \leq x, y \leq 1$, as it is given in (3.1). Let $0 \leq x_1 < x_2 \leq 1$, $0 \leq y_1 < y_2 \leq 1$. Then we have

$$\begin{aligned}
 & f(x_1, y_1)f(x_2, y_2) - f(x_2, y_1)f(x_1, y_2) \\
 = & [1 + \alpha A(x_1)A(y_1)][1 + \alpha A(x_2)A(y_2)] \\
 & - [1 + \alpha A(x_1)A(y_2)][1 + \alpha A(x_2)A(y_1)] \\
 = & \alpha A(x_2)A(y_2) + \alpha A(x_1)A(y_1) - \alpha A(x_1)A(y_2) - \alpha A(x_2)A(y_1) \\
 = & \alpha [A(y_2) - A(y_1)][A(x_2) - A(x_1)].
 \end{aligned} \tag{3.6}$$

It follows from (3.6) that if $A(x)$ is a monotone function of x , then equation (3.1) is TP_2 for $\alpha \geq 0$.

Let for example $A(x) = \varphi(x) - \varphi(1-x)$, where $\varphi(x) = e^{-nx}x^p$, $p > 1$, $n \geq 0$ as it is given in Subsection 3.1. Since $A(x)$ increases in $[0, 1]$ if $p \geq n$ then equation (3.1) with $\varphi(x) = e^{-nx}x^p$ is TP_2 for $p \geq n$ and $\alpha \geq 0$. Similarly, if $\varphi(x) = x^pe^{-nx^q}$, $A(x) = x^pe^{-nx^q} - (1-x)^pe^{-n(1-x)^q}$ as given in Subsection 3.2 it can be verified that $A'(x) = x^{p-1}e^{-nx^q}(p - nqx^q) + (1-x)^{p-1}e^{-n(1-x)^q}(p - nq(1-x)^q)$. Therefore $A(x)$, $0 \leq x \leq 1$ increases if $p \geq nq$, i.e. equation (3.1) with $\varphi(x) = x^pe^{-nx^q}$ is TP_2 for $p \geq nq$ and $\alpha \geq 0$.

4. Example for generalized Sarmanov bivariate distributions

In this section we construct a class of bivariate distributions motivated by the considerations provided in the previous section. This class is constructed by utilizing the general form (2.1), but it does not quite fit the Sarmanov construction. Its structure involves the independent part (the product of marginal densities) times a "correction factor" of the form $\{1 + \alpha H(F(x), G(y))\}$, essentially due to Farlie (1960). This form possesses attractive statistical properties in addition to its transparent probabilistic interpretation. It is thus an appropriate and flexible candidate for models in various fields including survival distributions. It will be shown that maximal correlation coefficients (positive and negative) are exceeding 0.5 in both negative and positive directions and can be increased by means of an auxiliary procedure.

Example 4.1. Let $0 < x_0 < 1/2 < x_1 < 1$, where $x_1 = 1 - x_0$. Assume that n and p are as in Example 3.2. Denote

$$a = e^{-nx_0} x_0^p (1-x_0)^p e^{-n(1-x_0)}, \quad \varphi_1(x) = \frac{ax_0 - 2ax}{x_0}, \quad \varphi_3(x) = a \left(1 - \frac{2(x-x_1)}{1-x_1} \right)$$

and

$$\varphi_2(x) = e^{-nx} x^p.$$

Consider the family of bivariate distributions with uniform marginals, given by p.d.f. of the form

$$f_\alpha(x, y) = 1 + \alpha H(x, y), \quad (4.1)$$

where

$$H(x, y) = \begin{cases} \varphi_1(x)\varphi_1(y) & \text{if } 0 \leq x, y \leq x_0 \\ (\varphi_2(x) - \varphi_2(1-x))(\varphi_2(y) - \varphi_2(1-y)) & \text{if } x_0 < x, y \leq x_1 \\ \varphi_3(x)\varphi_3(y) & \text{if } x_1 < x, y \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad (4.2)$$

Since $\int_0^{x_0} \varphi_1(x) dx = \int_{x_1}^1 \varphi_3(x) dx = \int_{x_0}^{x_1} (\varphi_2(x) - \varphi_2(1-x)) dx = 0$, we have $\int_0^1 H(x, y) dx = 0$, $\int_0^1 H(x, y) dy = 0$. It is clear that

$$-\frac{1}{\max_{x, y \in A^+} H(x, y)} \leq \alpha \leq \frac{1}{\max_{x, y \in A^-} (-H(x, y))},$$

where A^+ and A^- are subsets of $[0, 1]$, $H(x, y)$ takes on positive and negative values, respectively. For the family (4.1) the correlation coefficient

can be calculated in the usual manner yielding

$$\begin{aligned} \rho &= \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = 12\alpha \int_0^1 \int_0^1 xy H(x, y) dx dy \\ &= 12\alpha \left\{ \left[\int_0^{x_0} x \varphi_1(x) dx \right]^2 + \left[\int_{x_0}^{x_1} x (\varphi_2(x) - \varphi_2(1-x)) dx \right]^2 + \left[\int_{x_1}^1 x \varphi_3(x) dx \right]^2 \right\}. \end{aligned} \quad (4.3)$$

Let $n = 5.2$, $p = 3$. $x_0 = 0.001$, $x_1 = 0.999$. Then by the symmetry of construction we have $\max_{x, y \in A^+} H(x, y) = \max_{x, y \in A^-} (-H(x, y)) = \alpha^* = 0.006208$. That is, the admissible range for α is

$$-0.0062^{-2} \leq \alpha \leq 0.0062^{-2} \quad (-26014.6 \leq \alpha \leq 26014.6).$$

From (4.3) we have

$$\begin{aligned} 12\alpha_{\max} \left[\int_{x_0}^{x_1} x (\varphi_2(x) - \varphi_2(1-x)) dx \right]^2 &= 12 \times 0.0062^{-2} \times \\ \times \left\{ \int_{0.001}^{0.999} x \left[e^{-5.2x} x^3 - e^{-5.2(1-x)} (1-x)^3 \right] dx \right\}^2 &= 0.512. \end{aligned} \quad (4.4)$$

Consequently $\rho_{\max} > 0.512$.

We have thus presented the natural avenue for generating bivariate distributions of the FGM structure resulting in a high degree of correlation between the components. Obviously the results can be improved by a judicious choice of the functions.

5. Extensions to applied statistics

The insurance applications of risk theory entails the study of aggregate claim amounts (risks) for such purposes as ruin probability calculations and risk ordering. Underlying processes that govern the arrival and size of claims are usually contingent on some events like the occurrence of earthquakes in a location where insured individuals reside. Hence, dependence among the underlying claim processes exists. Some recent papers in the risk and insurance analyses depict these kinds of situations (see Hu and Wu, 1999, Dhaene and Denuit, 1999). Dependence between aggregate claim sizes X and Y can be measured by correlation measures as shown in our work. The ordering of risks by correlation mode is a natural extension of the concordance in the bivariate distribution case.

In the reliability analysis, one usually deals with systems whose sub-systems or components show dependent failures and survivals due to multiple effect shock contingencies. In these cases, distributional modelling about lifetime, hazard due to replacement and similar other magnitudes need to be used (see Hoyland and Rausand, 1994, Barlow and Proschan, 1981). It is either for the distributional properties, or the dependence measures or both that one can utilize the new general class of distributions that we introduce here for analysis and interpretation.

Further uses of FGM and Sarmanov multivariate modelling can be extended to attempts where more precise explanation of dependency is essential. The variety of application areas is inherent in many data analysis cases (see Cox and Wermuth, 1996, Joe, 1997).

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